

# LATER NEGATIVE ANSWERS TO TWO GUARDED-FRAISSE QUESTIONS IN ARXIV:2408.12755

FA BANACH RUN PACKET

## STATUS

This packet records a later-literature answer to two questions from Cuth–de Rancourt–Doucha, *Guarded Fraisse Banach spaces*, arXiv:2408.12755. In Section 7 the source asks:

Is every guarded Fraisse, respectively cofinally Fraisse, Banach space  $\omega$ -categorical?  
Does every guarded Fraisse, respectively cofinally Fraisse, Banach space have a closed age?

The later paper of de Rancourt–Fakhoury, *Isometric rigidity and Fraisse properties of Orlicz sequence spaces*, arXiv:2604.02080, answers both questions negatively, even in the cofinally Fraisse class.

## ANSWERING THEOREM

Let  $p > 3$ , and define the Orlicz function

$$M(t) = t^p e^{t^{-1}}.$$

Then the Orlicz sequence space  $\ell_M$  is cofinally Fraisse. Moreover,  $\ell_M$  is not oligomorphic, does not contain an isomorphic copy of  $\ell_2$ , and has non-closed age.

## IMPLICATIONS FOR ARXIV:2408.12755

The source paper uses  $\omega$ -categoricity in the continuous-logic sense, equivalently oligomorphy of the isometry action on finite powers of the unit ball. Therefore “not oligomorphic” is precisely a negative answer to the  $\omega$ -categoricity question.

The answering paper proves that  $\ell_M$  is cofinally Fraisse, and cofinally Fraisse spaces are guarded Fraisse. Since this same space is not oligomorphic, Question `quest:GFImpliesOmegaCat` has a negative answer for both “guarded” and “cofinally guarded” versions.

The answering theorem also states that  $\ell_M$  has non-closed age. Since  $\ell_M$  is cofinally Fraisse, Question `quest:closedAge` likewise has a negative answer for both the guarded and cofinally Fraisse versions.

Finally, the source’s broad request for new infinite-dimensional guarded or cofinally Fraisse Banach spaces is answered positively by this Orlicz sequence family. This does not settle the source’s more specific subquestions about  $L_p(L_q(L_r))$  or  $L_p(E)$ .

## PROVENANCE

The cheap run indexes contained no pre-existing promoted packet for this answer when this note was created. The relevant source facts are:

- arXiv:2408.12755, Section 7, labels `quest:GFImpliesOmegaCat` and `quest:closedAge`.
- arXiv:2604.02080, abstract: the authors prove guarded Fraisse Orlicz sequence spaces that are not  $\omega$ -categorical and have non-closed age, and say this answers a question of Cuth–de Rancourt–Doucha.

- arXiv:2604.02080, Corollary `cor:OrliczFraisie` and Theorem `thm:Counterexample`:  $M(t) = t^p e^{t-1}$ ,  $p > 3$ , gives the stated cofinally Fraisse counterexample.

## REFERENCES

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- [1] M. Cuth, N. de Rancourt, and M. Doucha, *Guarded Fraisse Banach spaces*, arXiv:2408.12755.
- [2] N. de Rancourt and M. Fakhoury, *Isometric rigidity and Fraisse properties of Orlicz sequence spaces*, arXiv:2604.02080.